14.6.1 Functions of constructor

As we know now that the constructor is a special initialization member function of a class that is called automatically whenever an instance of a class is declared or created. The main function of the constructor is

1) To allocate memory space to the object and

2) To initialize the data member of the class object

There is an alternate way to initialize the class objects but in that case we have to explicitly call the member function.

14.7 Types of constructors

There are different types of constructors.

- **Default Constructors**

A constructor that accepts no parameter is called default constructor. For example in the class Data, Data ::Data() is the default constructor . Using this constructor Objects are created similar to the way the variables of other data types are created. If a class does not contain an explicit constructor (user defined constructor) the compiler automatically generate a default constructor.

- **Parameterized Constructors**

A constructor which can take arguments is called parameterized constructor .This type of constructor helps to create objects with different initial values. This is achieved by passing parameters to the function.

Example :

```
Data :: Data(int,int);
```

- **Copy Constructors**

A constructor having a reference to an already existing object of its own class is called copy constructor. It is usually of the form Data (Data&), where Data is the class name.

A copy constructor can be called in many ways:

1) When an object is passed as a parameter to any of the member functions

   Example void Data::putdata(Data x);

2) When a member function returns an object

   Example Data getdata() { }

3) When an object is passed by reference to an instance of its own class

   For example, Data d1, d2 (d1); // d2(d1) calls copy constructor

11th Std Computer Science_EM Chapter 14.indd 241 06-01-2023 15:11:50